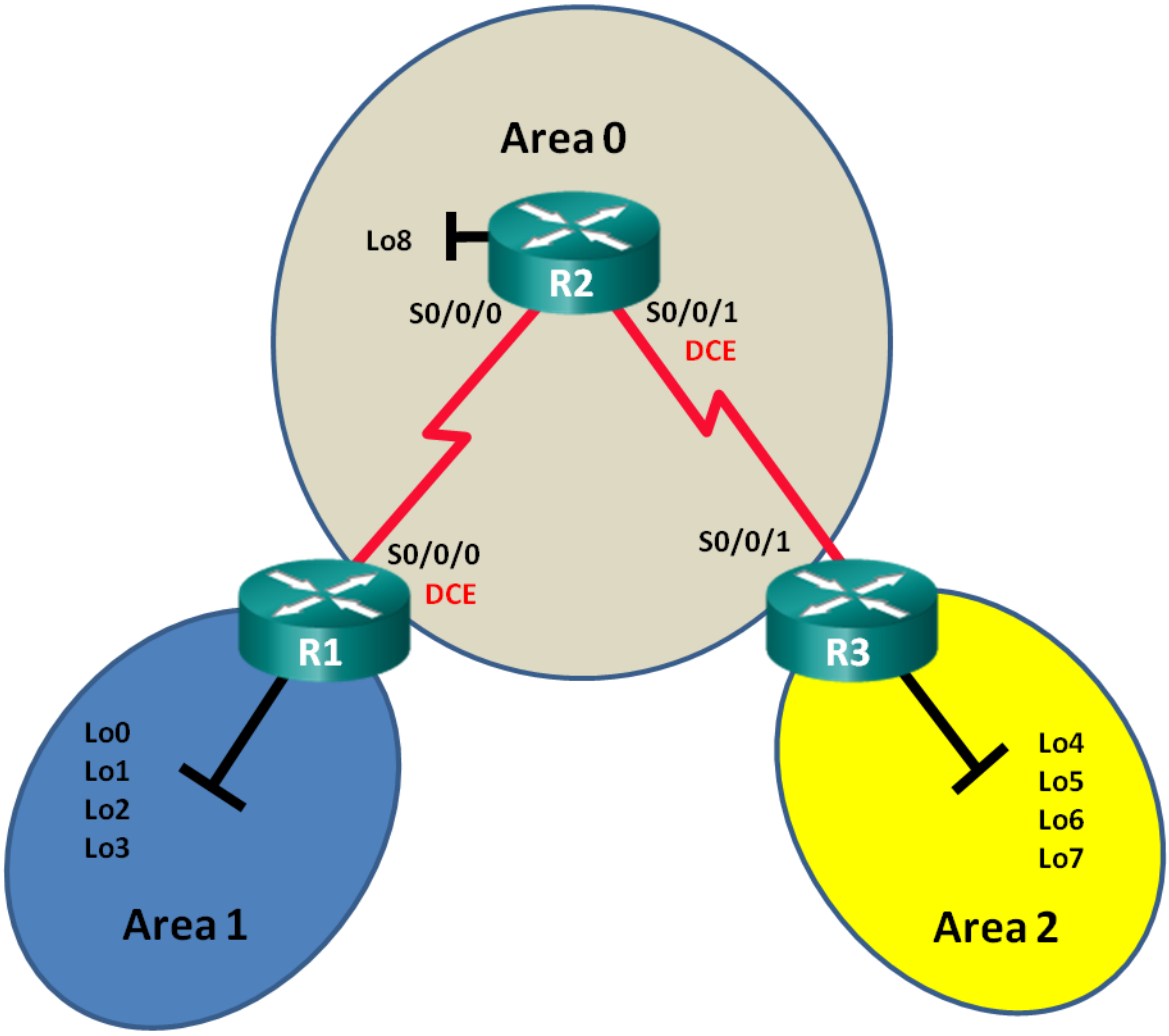
Lab - Configuring Multi-area OSPFv3

Topology



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Addressing Table

Device	Interface	IPv6 Address
R1	S0/0/0 (DCE)	2001:DB8:ACAD:12::1/64 FE80::1 link-local
	Lo0	2001:DB8:ACAD::1/64
	Lo1	2001:DB8:ACAD:1::1/64
	Lo2	2001:DB8:ACAD:2::1/64
	Lo3	2001:DB8:ACAD:3::1/64
R2	S0/0/0	2001:DB8:ACAD:12::2/64 FE80::2 link-local
	S0/0/1 (DCE)	2001:DB8:ACAD:23::2/64 FE80::2 link-local
	Lo8	2001:DB8:ACAD:8::1/64
R3	S0/0/1	2001:DB8:ACAD:23::3/64 FE80::3 link-local
	Lo4	2001:DB8:ACAD:4::1/64
	Lo5	2001:DB8:ACAD:5::1/64
	Lo6	2001:DB8:ACAD:6::1/64
	Lo7	2001:DB8:ACAD:7::1/64

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Configure Multi-area OSPFv3 Routing

Background / Scenario

Using multi-area OSPFv3 in large IPv6 network deployments can reduce router processing cycles by creating smaller routing tables and requiring less memory overhead. In multi-area OSPFv3, all areas are connected to the backbone area (area 0) through area border routers (ABRs).

In this lab, you will implement OSPFv3 routing for multiple areas. You will also use a number of **show** commands to display and verify OSPFv3 routing information. This lab uses loopback addresses to simulate networks in multiple OSPFv3 areas.

Note: The routers used with CCNA hands-on labs are Cisco 1941 Integrated Services Routers (ISRs) with Cisco IOS Release 15.2(4)M3 (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at this end of this lab for the correct interface identifiers.

Note: Make sure that the routers have been erased and have no startup configurations. If you are unsure, contact your instructor.

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Required Resources

- 3 Routers (Cisco 1941 with Cisco IOS Release 15.2(4)M3 universal image or comparable)
- 3 PCs (Windows with terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Serial cables as shown in the topology

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings on the routers.

Step 1: Cable the network as shown in the topology.

Step 2: Initialize and reload the routers as necessary.

Step 3: Configure basic settings for each router.

a. Disable DNS lookup.
b. Configure device name as shown in the topology.
c. Assign **class** as the privileged EXEC password.
d. Assign **cisco** as the vty password.
e. Configure a MOTD banner to warn users that unauthorized access is prohibited.
f. Configure **logging synchronous** for the console line.
g. Encrypt plaintext passwords.
h. Configure the IPv6 unicast and link-local addresses listed in the Addressing Table for all interfaces.
i. Enable IPv6 unicast routing on each router.
j. Copy the running configuration to the startup configuration.

Step 4: Test connectivity.

The routers should be able to ping one another. The routers are unable to ping distant loopbacks until OSPFv3 routing is configured. Verify and troubleshoot if necessary.

Part 2: Configure Multi-area OSPFv3 Routing

In Part 2, you will configure OSPFv3 routing on all routers to separate the network domain into three distinct areas, and then verify that routing tables are updated correctly.

Step 1: Assign router IDs.

a. On R1, issue the **ipv6 router ospf** command to start an OSPFv3 process on the router.

```
R1(config)# ipv6 router ospf 1
```

Note: The OSPF process ID is kept locally and has no meaning to other routers on the network.

b. Assign the OSPFv3 router ID **1.1.1.1** to R1.

```
R1(config-rtr)# router-id 1.1.1.1
```

c. Start an OSPFv3 process on R2 and R3 and assign a router ID of **2.2.2.2** to R2 and a router ID of **3.3.3.3** to R3.

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d. Issue the **show ipv6 ospf** command to verify the router IDs on all routers.

```
R2# show ipv6 ospf
 Routing Process "ospfv3 1" with ID 2.2.2.2
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 Router is not originating router-LSAs with maximum metric
 <output omitted>
```

Step 2: Configure multi-area OSPFv3.

a. Issue the **ipv6 ospf 1 area** *area-id* command for each interface on R1 that is to participate in OSPFv3 routing. The loopback interfaces are assigned to area 1 and the serial interface is assigned to area 0. You will change the network type on the loopback interfaces to ensure that the correct subnet is advertised.

```
R1(config)# interface lo0
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)# interface lo1
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)# interface lo2
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)# interface lo3
R1(config-if)# ipv6 ospf 1 area 1
R1(config-if)# ipv6 ospf network point-to-point
R1(config-if)# interface s0/0/0
R1(config-if)# ipv6 ospf 1 area 0
```

b. Use the **show ipv6 protocols** command to verify multi-area OSPFv3 status.

```
R1# show ipv6 protocols
IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 1"
  Router ID 1.1.1.1
  Area border router
  Number of areas: 2 normal, 0 stub, 0 nssa
  Interfaces (Area 0):
    Serial0/0/0
  Interfaces (Area 1):
    Loopback0
    Loopback1
    Loopback2
    Loopback3
  Redistribution:
    None
```

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c. Assign all interfaces on R2 to participate in OSPFv3 area 0. For the loopback interface, change the network type to point-to point. Write the commands used in the space below.

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d. Use the **show ipv6 ospf interface brief** command to view OSPFv3-enabled interfaces.

```
R2# show ipv6 ospf interface brief
Interface     PID   Area            Intf ID    Cost  State Nbrs F/C
Lo8           1     0               13         1     P2P   0/0
Se0/0/1       1     0               7          64    P2P   1/1
Se0/0/0       1     0               6          64    P2P   1/1
```

e. Assign the loopback interfaces on R3 to participate in OSPFv3 area 2 and change the network type to point-to-point. Assign the serial interface to participate in OSPFv3 area 0. Write the commands used in the space below.

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f. Use the **show ipv6 ospf** command to verify configurations.

```
R3# show ipv6 ospf
 Routing Process "ospfv3 1" with ID 3.3.3.3
 Event-log enabled, Maximum number of events: 1000, Mode: cyclic
 It is an area border router
 Router is not originating router-LSAs with maximum metric
 Initial SPF schedule delay 5000 msecs
 Minimum hold time between two consecutive SPFs 10000 msecs
 Maximum wait time between two consecutive SPFs 10000 msecs
 Minimum LSA interval 5 secs
 Minimum LSA arrival 1000 msecs
 LSA group pacing timer 240 secs
 Interface flood pacing timer 33 msecs
 Retransmission pacing timer 66 msecs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of areas in this router is 2. 2 normal 0 stub 0 nssa
 Graceful restart helper support enabled
```

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```
 Reference bandwidth unit is 100 mbps
 RFC1583 compatibility enabled
    Area BACKBONE(0)
        Number of interfaces in this area is 1
        SPF algorithm executed 2 times
        Number of LSA 16. Checksum Sum 0x0929F8
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
    Area 2
        Number of interfaces in this area is 4
        SPF algorithm executed 2 times
        Number of LSA 13. Checksum Sum 0x048E3C
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```

Step 3: Verify OSPFv3 neighbors and routing information.

a. Issue the **show ipv6 ospf neighbor** command on all routers to verify that each router is listing the correct routers as neighbors.

```
R1# show ipv6 ospf neighbor

            OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
2.2.2.2           0   FULL/  -        00:00:39    6               Serial0/0/0
```

b. Issue the **show ipv6 route ospf** command on all routers to verify that each router has learned routes to all networks in the Addressing Table.

```
R1# show ipv6 route ospf
IPv6 Routing Table - default - 16 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
       I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
       EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
       NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
       OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
OI  2001:DB8:ACAD:4::/64 [110/129]
     via FE80::2, Serial0/0/0
OI  2001:DB8:ACAD:5::/64 [110/129]
     via FE80::2, Serial0/0/0
OI  2001:DB8:ACAD:6::/64 [110/129]
     via FE80::2, Serial0/0/0
OI  2001:DB8:ACAD:7::/64 [110/129]
     via FE80::2, Serial0/0/0
O   2001:DB8:ACAD:8::/64 [110/65]
     via FE80::2, Serial0/0/0
```

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```
O   2001:DB8:ACAD:23::/64 [110/128]
     via FE80::2, Serial0/0/0
```

What does OI stand for? How was the OI route learned?

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c. Issue the **show ipv6 ospf database** command on all routers.

```
R1# show ipv6 ospf database

            OSPFv3 Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

ADV Router       Age         Seq#        Fragment ID  Link count  Bits
 1.1.1.1         908         0x80000001  0            1           B
 2.2.2.2         898         0x80000003  0            2           None
 3.3.3.3         899         0x80000001  0            1           B

                Inter Area Prefix Link States (Area 0)

ADV Router       Age         Seq#        Prefix
 1.1.1.1         907         0x80000001  2001:DB8:ACAD::/62
 3.3.3.3         898         0x80000001  2001:DB8:ACAD:4::/62

                Link (Type-8) Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Interface
 1.1.1.1         908         0x80000001  6          Se0/0/0
 2.2.2.2         909         0x80000002  6          Se0/0/0

                Intra Area Prefix Link States (Area 0)

ADV Router       Age         Seq#        Link ID    Ref-lstype  Ref-LSID
 1.1.1.1         908         0x80000001  0          0x2001      0
 2.2.2.2         898         0x80000003  0          0x2001      0
 3.3.3.3         899         0x80000001  0          0x2001      0

                Router Link States (Area 1)

ADV Router       Age         Seq#        Fragment ID  Link count  Bits
 1.1.1.1         908         0x80000001  0            0           B

                Inter Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Prefix
 1.1.1.1         907         0x80000001  2001:DB8:ACAD:12::/64
 1.1.1.1         907         0x80000001  2001:DB8:ACAD:8::/64
 1.1.1.1         888         0x80000001  2001:DB8:ACAD:23::/64
 1.1.1.1         888         0x80000001  2001:DB8:ACAD:4::/62
```

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```
                Link (Type-8) Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Interface
 1.1.1.1         908         0x80000001  13         Lo0
 1.1.1.1         908         0x80000001  14         Lo1
 1.1.1.1         908         0x80000001  15         Lo2
 1.1.1.1         908         0x80000001  16         Lo3

                Intra Area Prefix Link States (Area 1)

ADV Router       Age         Seq#        Link ID    Ref-lstype  Ref-LSID
 1.1.1.1         908         0x80000001  0          0x2001      0
```

How many link state databases are found on R1? _____

How many link state databases are found on R2? _____

How many link state databases are found on R3? _____

Reflection

1. Why would multi-area OSPFv3 be used?

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